

Expected-Return Inference over Complete Policies

Eden Havri

Ronen Brafman

Ben-Gurion University of the Negev

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Abstract

Inference over complete policies distinguishes uncertainty in selecting a decision rule from randomness in the environment. We study a Gibbs distribution that weights deterministic policies by expected return, rather than by exponentiated trajectory outcomes. Building on established policy-inference formulations, we develop a simulator-based mean-field variational implementation. A score-function gradient combines unbiased rollout returns, an independent baseline, and exact policy-space entropy; each sampled policy keeps its decisions on state revisits. We establish the gradient’s unbiasedness under explicit assumptions and validate the implementation by exact enumeration on a small stochastic MDP. The evaluation design separates target specification, approximation error, deployment semantics, and control quality, charging exploration and refinement to a common interaction allowance. The confirmatory all-method study is not yet complete; this draft makes no empirical competitiveness or superiority claim.

1 Introduction

A successful trajectory need not identify a good decision rule. Stochastic transitions can make the same policy succeed on one attempt and fail on another. Inference-based control must therefore specify whether it favors fortunate trajectories or policies with high expected return. A second choice concerns execution: selecting an entire policy once is generally different from independently resampling its actions on every state visit.

We investigate these choices through distributions over complete stationary deterministic policies. A sampled policy fixes an action at every decision state and retains that action on revisits. The distribution weights policies by the exponential of their expected return. This starting point has direct precedent: [Wingate et al. \(2011\)](#) combine a policy prior, expected-value exponential, and policy-space Metropolis–Hastings. Information-cost approaches ([Ortega and Braun, 2011](#)) and the broader control-as-inference literature ([Levine, 2018](#)) provide related interpretations. The Gibbs distribution itself is not claimed as a new principle.

The practical question is how to optimize a tractable policy distribution using only a simulator, while preserving the expected-return objective and the decision rule being evaluated. An unbiased return estimate does not become an unbiased policy weight after exponentiation. Conversely, a restricted variational family can support useful control without representing every posterior dependence. Both distinctions matter when judging a practical inference-based learner.

Our implementation takes the variational route: it learns a factorized distribution over complete policies using a linear expected-return gradient and exact policy-space entropy. A separately costed PPO initializer supplies exploration, not the reference measure. The methodological contribution is an explicitly validated simulator gradient for this policy-table formulation, together with an execution-aware evaluation contract. It does not establish a new Gibbs principle, global convergence, or calibrated uncertainty. The planned control study

compares all declared methods at a matched total interaction allowance. Its held-out results are not yet available and are not replaced here by development scores.

2 Target and decision rule

Consider a finite-horizon stochastic MDP with bounded trajectory return G_H , decision states $\mathcal{D}$, and a finite class Π of stationary deterministic policies. Write $J(\pi) = \mathbb{E}[G_H \mid \pi]$. For a full-support reference μ and inverse temperature $\beta > 0$, define

$$P_\beta(\pi) = \frac{\mu(\pi)e^{\beta J(\pi)}}{Z_\beta}, \quad Z_\beta = \sum_{\pi \in \Pi} \mu(\pi)e^{\beta J(\pi)}. \quad (1)$$

We use uniform μ , so its modes maximize expected return within Π . Finite-temperature samples need not be optimal controllers. This generalized posterior describes preference-weighted policy selection, not calibrated epistemic uncertainty about unknown dynamics.

Expectation precedes exponentiation. In general, $e^{\beta \mathbb{E}[G_H \mid \pi]} \neq \mathbb{E}[e^{\beta G_H} \mid \pi]$. For example, policies returning zero surely or ± 1 equiprobably have equal expected return. Equation (1) assigns them equal mass under a uniform reference, whereas exponential trajectory weighting favors the variable policy by a factor $\cosh(\beta)$. This is a change of objective, not merely an imperfect estimate of the intended weight.

Commitment and action resampling. A draw $\pi \sim Q$ followed throughout an episode gives value $\mathbb{E}_Q[J(\pi)]$. Executing Q 's statewise action marginals afresh on every visit generally gives a different value. Even a factorized policy distribution preserves repeated-visit consistency under commitment; marginal execution does not. The evaluation contract reports both rules, using native stochastic execution for practical comparisons with stochastic RL baselines.

Stationarity is an additional restriction: the finite-horizon optimum may depend on time-to-go. Dynamic programming over time supplies an upper control, not a same-class stationary competitor. Reward rescaling also requires reciprocal rescaling of β ; temperature is part of the target specification, not a parameter selected on held-out outcomes.

3 Direct variational policy inference

Let $q_\theta(\pi) = \prod_{s \in \mathcal{D}} q_{\theta,s}(\pi(s))$ be a categorical mean-field distribution. Its objective is

$$\mathcal{L}(\theta) = \beta \mathbb{E}_{q_\theta}[J(\pi)] - \text{KL}(q_\theta \parallel \mu), \quad (2)$$

$$\log Z_\beta - \mathcal{L}(\theta) = \text{KL}(q_\theta \parallel P_\beta). \quad (3)$$

The identity separates optimization from approximation: a well-optimized factorized family can still miss policy correlations. Under uniform μ , the negative KL equals the sum of categorical entropies minus $|\mathcal{D}| \log |\mathcal{A}|$. This is entropy over policy tables, not visitation-weighted action entropy.

Simulator gradient. Draw N independent policies and K independent rollouts per policy. Set $\bar{G}_i = K^{-1} \sum_k G_H(\pi_i, \xi_{ik})$ and, for $N > 1$, $b_i = \beta(N-1)^{-1} \sum_{j \neq i} \bar{G}_j$. Let M_{is} indicate that policy i 's action at s is used before the horizon in at least one rollout, not merely that its final state is s . The implemented gradient estimator is

$$\hat{g} = \frac{1}{N} \sum_{i=1}^N (\beta \bar{G}_i - b_i) \sum_{s \in \mathcal{D}} M_{is} \nabla_\theta \log q_{\theta,s}(\pi_i(s)) - \nabla_\theta \text{KL}(q_\theta \parallel \mu). \quad (4)$$

For a singleton minibatch, use $b_i = 0$. The return and baseline are detached in the score-function surrogate (Williams, 1992); the KL term is differentiated exactly.

To justify the mask, condition on all other policy coordinates and simulator randomness. Whether s is first visited cannot depend on $\pi(s)$, which has not yet been used. On the event of no visit, the return is independent of that action, and the omitted score contribution has expectation zero. The masked score itself has zero mean. The leave-one-out baseline is independent of policy i and its rollouts, so it also contributes zero in expectation. Consequently, Equation (4) is unbiased under the declared independent mean-field sampling scheme. Each policy coordinate is scored once, not once per visit.

The mask also removes variance from unused policy coordinates. At fixed parameters, conditional expectation of the otherwise identical full-table score given the observed trajectories and used actions equals the masked score. The law of total covariance therefore gives a positive-semidefinite covariance reduction. This standard conditional-expectation argument compares the same returns and baseline; it guarantees neither faster learning nor better control than other estimators or competitors.

Because the return term is linear, $K = 1$ is valid: an unbiased rollout return suffices for an unbiased gradient. Larger K trades lower conditional rollout variance against fewer policy draws or updates at a fixed interaction budget. This argument would not justify exponentiating one noisy return as an exact inference weight.

Initialization and computation. PPO-Clip (Schulman et al., 2017) first trains a statewise guide. During this stage, the current candidate adds $1/\sqrt{n(s')}$ to each transition’s training reward, where $n(s')$ is the incremented count of arrivals at its next state s' . Refinement and evaluation use only the original rewards. Its clipped, positive action probabilities initialize the logits; they do not replace μ . Adam then updates the variational logits using original environment rewards and Equation (4). Fresh policies are sampled at each update, with no persistent particle genealogy or resampling stage. The frozen confirmation configuration uses $N = 8$, $K = 1$, $\beta = 64$, and learning rate 0.01, with half of the 200k-transition allowance reserved for initialization. These choices were selected on development data before protected outcomes; they are not universal constants.

One refinement update. Draw independent complete policy tables from the current categorical distribution, execute each table consistently for its independent rollouts, and record returns plus the union of visited states. Compute Equation (4), score each visited coordinate once, and update the logits. Discard the sampled tables and repeat with fresh draws. Exploration training is a distinct, charged initialization stage, not an unreported prerequisite or a source of persistent particles.

The total simulator allowance includes both initialization and refinement. Before sampling a minibatch, the runner reserves its worst-case NKH transitions, reducing N or stopping when insufficient allowance remains. Returns are not selectively truncated to fit a budget. Complete-episode baseline trainers receive a stopping threshold that reserves their maximum one-episode overshoot. Core baseline updates are unchanged, and actual consumed transitions and unused allowance are reported. Exact model evaluation is offline only; it never supplies a training score or gradient.

4 Correctness and evaluation design

Correctness and approximation are different checks. On small enumerable MDPs, exact differentiation of Equation (2) is compared with the expectation of the actual implementation. Enumerating policy pairs and stochastic visitation outcomes with $K = 1, 2$ and

the baseline enabled or disabled yields agreement to 10^{-12} on the diagnostic instance. This validates the estimator there, not global optimization or posterior equality on larger domains.

A separate four-policy MDP makes the family restriction explicit: three policies return 1 and the fourth returns 0. At $\beta = 64$, a certified grid search bounds the best mean-field reverse KL between 0.369 and 0.405 nats, even though its expected return and the exact target's are both essentially 1. The thesis gives the finite-grid error bound. High control return therefore does not certify an accurate target distribution.

Tasks and baselines. The planned main study uses stochastic GridWorld tiers with 3×3 , 5×5 , and 8×8 cells, increasing obstacle density, transition noise, and horizon. Navigation maps contain walls, gravel, dirt, terminal swamps, and a goal. A paired secondary panel adds repeatedly rewarding meadow cells: there, maximizing utility need not maximize goal-reaching probability. Confirmation uses ten fresh base geometries per tier and three independent training blocks per geometry: 30 independent base mazes and 90 paired blocks. The two terrain panels share geometry and training blocks; they are not independent replicates. Development maps are separate from confirmation maps. The comparison roster includes REINFORCE with a value baseline, PPO-Clip with Monte Carlo advantages, fixed-temperature discrete SAC, categorical CEM, and stationary-output Double Q-learning (Williams, 1992; Schulman et al., 2017; Christodoulou, 2019; Mannor et al., 2003; van Hasselt, 2010). These are precisely specified implementations, not a claim to cover every state-of-the-art learner. Pure inference, PPO-initialized inference, and continued exploration-assisted PPO separate inference from initialization and its exploration aid; initialization is not free additional training.

Metrics and execution. For each saved policy distribution, exact environment expectations evaluate committed table draws and per-visit action resampling separately. Monte Carlo error across table draws is conditional evaluation error, not uncertainty across training runs. Return and goal probability are distinct outcomes. Uniform random behavior and known-model, time-dependent dynamic programming provide visible controls; return-optimal and goal-optimal upper bounds are computed separately. Neither oracle is a simulator-only learned competitor.

Budget and attribution. The prospective design uses a bounded 100k-transition configuration screen followed by 200k-transition validation on fresh development maps; configuration rankings need not transfer between these budgets. The final allowance and inference variant are frozen before protected evaluation. All active training transitions count, including any PPO initializer; absorbing padding does not. Saved training policies are evaluated offline without consuming training randomness or choosing checkpoints from test returns. Learning curves report actual interactions, and cost reports distinguish training, evaluation, and allocated CPU-hours. All configurations receive the same predeclared 2,048-table terminal readout before selection; confirmation retains this precision. A separate 4,096-episode evaluation per learned run measures actual return tails, swamp termination, and timeout, rather than a distribution of policy expectations. Per-visit stochastic expected values remain exact. A full-budget assisted PPO control and cold-start VI distinguish exploration assistance from the complete inference pipeline. Differences between complete systems do not, by themselves, isolate objective or representation effects.

Confirmation and reporting. Candidate settings, temperature, tuning allowances, maps, seeds, exclusions, sample size, and analysis are frozen before held-out outcomes are inspected. Independent training runs and maps, not within-run policy draws, are the statistical units (Agarwal et al., 2021). Reporting first averages training blocks within each map, then weights maps equally within each of the six tier/panel strata. Descriptive 95% intervals use joint

paired resampling of base geometries followed by training blocks, retaining both panels and all methods together. Final tables and figures must include every declared core method, random and oracle controls, uncertainty, and valid unfavorable outcomes. Similar point estimates do not establish equivalence; any such claim needs a justified, predeclared practical margin.

Results status. The complete replicated held-out comparison is pending. No development or partially completed leaderboard is included in this submission-paper source. Final empirical claims await a complete independently verified study.

5 Discussion

Expected-return inference is a target specification; it is not tied to a particular sampler. Direct variational learning avoids noisy exponential weights and permits simulator-only refinement, but introduces a restricted family and stochastic optimization. Its policy-table entropy remains an explicit preference, not a guarantee of higher control return. A simulator gradient can be unbiased while providing little useful feedback in a sparse-reward environment; exploration remains a substantive requirement. The restricted family can also miss dependencies even when its expected return is high. The remaining empirical question is when this complete pipeline offers useful control and cost trade-offs against the declared competitors, under both deployment rules. Until the held-out study is complete, competitiveness and superiority remain unestablished. The thesis retains extended derivations, development diagnostics, and implementation details without treating them as confirmatory evidence.

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